25900546 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Core Course (DSC) - MG3DSCPHL202 - INTRODUCTION TO SYMBOLIC LOGIC

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short answer

Answer any **8** questions

Each question carries **2** marks

1. Define validity. [K] / [CO1]
2. Distinguish between informative and emotive function of language. [K] / [CO1]
3. Give two examples of propositions. [K] / [CO1]
4. If p is true and q is false, determine the value of p →q [K] / [CO2]
5. Give an example of a logical expression depicting constants and variables. [K] / [CO2]
6. Symbolise the expression ' Ravi is a boy and a student'. [U] / [CO2]
7. Give an example of two logically equivalent statements. [K] / [CO3]
8. Give two examples of contingent statement. [K] / [CO3]
9. Define contradiction with an example. [K] / [CO3]

10. State the justification for the following formal proof of validity [U] / [CO4]

A. B/ ∴ A. B

A

A V C

11. State the rule of inference [K] / [CO4]

p →q

~q

∴ ~p

12. What is the need for a formal proof method to prove validity? [U] / [CO4]

[2x8 = 16]

Part B

Short essay

Answer any **6** questions

Each question carries **5** marks

13. Examine the importance of symbols in distinguishing symbolic logic from traditional logic. [K] / [CO1]

14. Discuss the main differences between propositional logic and predicate logic . [K] / [CO1]

15. Write the truth table for material equivalence. [U] / [CO2]

16. If A and B are true and X and Y are false but the values of P and Q are unknown, what can be said about the truth values of the following: [U] / [CO2]

a. P. ~P

b. P V ~Y

17. Examine with shorter truth table method which of the following are tautologous: [K] / [CO3]

a. (p v q) →(p →~q)

b. (p →q) →(p v q)

18. Using truth table method, determine the validity or invalidity of the following [K] / [CO3]

If Ed wins the first prize, then Fred wins second prize, and if Fred wins second prize then George is disappointed. Either Ed wins first prize or George is disappointed. Therefore Fred does not win second prize.

19. Construct truth table and determine the validity or invalidity of the following [K] / [CO3]

a. p →q

∴ ~q →~p

b. p →q
p V q
∴ q

20. Can an argument have more than one specific form . Why? [K] / [CO3]

21. Distinguish between constructive dilemma and destructive dilemma [K] / [CO4]

22. Using truth table method determine whether the following are tautologous [K] / [CO3]
a. p →(p→q)
b. p →(p V q)
c. (p.p) →p

[5x6 = 30]

Part C

Essay

Answer any **2** questions

Each question carries **12** marks

23. Discuss the relationship between logic and language [K] / [CO1]

24. 'Compound statements and their truth values are related to their corresponding simple statements.'Elaborate [K] / [CO2]

25. Write atleast two argument forms for the following [K] / [CO3]

a. p →(q.r)
∴ ~(q.r) →~p

b. p →(q V r)
p →~q
∴ p V r

c. (p V q) → (p. q)
p.q
∴ p V q

26. [K] / [CO3]
Explain with suitable examples how an argument can have multiple argument forms

[12x2 = 24]